EXAM PROJECT

Bayesian Classification of Lyric vs Dramatic Voice Structure from Opera Timbre Features

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Overview

Data Source

Scientific Reports: ***"New objective timbre parameters for classification of voice type and Fach in professional opera singers"***

Questions

Given a **known voice type** and a set of **vocalizations** from an unseen singer:

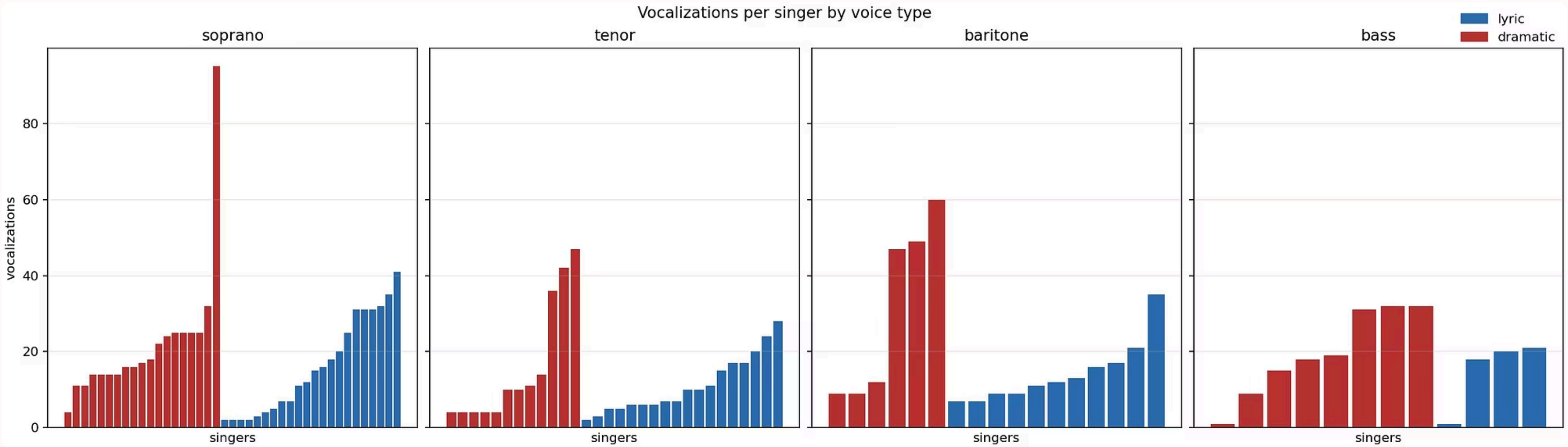
- What is the posterior probability of lyric vs dramatic voice structure?
- How does uncertainty change as more vocalizations are observed?

Data Summary

Voice type	Class	Vocalizations	Singers
Soprano	dramatic	422	19
Soprano	lyric	352	22
Tenor	dramatic	190	12
Tenor	lyric	199	18
Baritone	dramatic	186	6
Baritone	lyric	157	11
Bass	dramatic	157	8
Bass	lyric	60	4

Vocalizations

Vocalizations available are repeated measurements nested within singers. Each vocalization is associated with acoustic features provided by the paper's authors



Each bar is one singer; bar height is the number of available vocalizations

Training

Train-test split: all vocalizations from the same singer stay in the same split.

Features: PHE and SC

PHE (Position of Half Energy)

Normalized position of the half-energy frequency within the voice-type-specific band

SC (Spectral Centroid)

Center of mass of the energy distribution within the voice-type-specific frequency band

Sample ID	Voice type	Class	Vowel	Pitch	PHE	SC
S2-w001-Barb-A-as2-1	soprano	lyric	A	a2	**45.93**	**3264.9**
S1-w005-BrWa-A-c2-1	soprano	dramatic	A	c2	**28.57**	**3055.1**
T2-m014-Tami-A-g1-1	tenor	lyric	A	g1	**44.46**	**2713.5**
T1-m004-Flor-A-g1-1	tenor	dramatic	A	g1	**30.21**	**2616.0**

Normalization

PHE and SC are standardized before fitting the model

Bayesian Hierarchical Generative Model

Input

$$\mathbf{x}_{sj} = \begin{bmatrix} \mathrm{PHE}_{sj} \\ \mathrm{SC}_{sj} \end{bmatrix} \in \mathbb{R}^2$$

Each vocalization is represented by standardized PHE and SC. Singer s has repeated vocalizations $j = 1, ..., J$

$$v_s \in \{\text{soprano}, \text{tenor}, \text{baritone}, \text{bass}\}$$

$$c_s \in \{0, 1\}, \qquad 0 = \text{lyric}, \quad 1 = \text{dramatic}$$

Model parameters

$$\Theta = \left\{\pi_v, \boldsymbol{\mu}_{v,c}, \boldsymbol{\tau}_{v,c}, \boldsymbol{\sigma}_v : v \in \mathcal{V},\ c \in \{0,1\}\right\}$$

- π_v: dramatic-class prevalence for voice type v
- $\boldsymbol{\mu}_{v,c}$: group acoustic center
- $\boldsymbol{\tau}_{v,c}$: variability among singers within the same voice type and class
- $\boldsymbol{\sigma}_v$: vocalization to vocalization noise

Priors

$$\pi_v \sim \mathrm{Beta}(1, 1)$$

$$\boldsymbol{\mu}_{v,c} \sim \mathcal{N}\left(\mathbf{0}, 1.5^2 I_2\right)$$

$$\boldsymbol{\tau}_{v,c} \sim \mathrm{HalfNormal}(1)$$

$$\boldsymbol{\sigma}_v \sim \mathrm{HalfNormal}(1).$$

Generative story

$$c_s \mid v_s, \Theta \sim \mathrm{Bernoulli}\left(\pi_{v_s}\right) \qquad \text{singer class}$$

$$\mathbf{z}_s \mid c_s, v_s, \Theta \sim \mathcal{N}\left(\boldsymbol{\mu}_{v_s,c_s}, \mathrm{diag}\left(\boldsymbol{\tau}^2_{v_s,c_s}\right)\right) \qquad \text{latent singer signature}$$

$$\mathbf{x}_{sj} \mid \mathbf{z}_s, v_s, \Theta \sim \mathcal{N}\left(\mathbf{z}_s, \mathrm{diag}\left(\boldsymbol{\sigma}^2_{v_s}\right)\right) \qquad \text{observed vocalizations}$$

Bayesian Hierarchical Generative Model

Class Prior

$$p(c_* = c \mid \pi_{v_*})$$

Acoustic Evidence

$$p(\mathbf{X}_* \mid c, v_*, \Theta) = \int p(\mathbf{z}_* \mid c, v_*, \Theta) \prod_{j=1}^{J_*} p(\mathbf{x}_{*j} \mid \mathbf{z}_*, v_*, \Theta)\, d\mathbf{z}_*$$

Training Posterior

$$p(\Theta, \mathbf{z} \mid D_{\text{train}})$$

Posterior Averaging

$$S_c = \int p(c \mid \pi_{v_*}) p(\mathbf{X}_* \mid c, v_*, \Theta) p(\Theta \mid D_{\text{train}})\, d\Theta$$

Posterior Class Probability

$$p(c_* = \text{dramatic} \mid \mathbf{X}_*, v_*, D_{\text{train}}) = \frac{S_{\text{dramatic}}}{S_{\text{lyric}} + S_{\text{dramatic}}}$$

What the prediction combines

— **Class Prior**

How plausible is a class for that voice type?

— **Acoustic Evidence**

Compatibility of the observed vocalizations with a candidate class, after integrating out the singer's latent acoustic signature.

— **Posterior Averaging**

Prediction obtained by averaging class scores over the posterior distribution of the model parameters

NUTS Sampler

Why

The model defines a posterior distribution over unknown parameters and latent singer signatures. This posterior is too complex to compute analytically, so we approximate it with MCMC sampling.

What

- It is an adaptive Hamiltonian Monte Carlo algorithm.
- It uses gradients of the log posterior to explore high-dimensional continuous parameter spaces efficiently.

We sample from

$$p(\Theta, z \mid D_{\text{train}})$$

Θ contains model parameters: class prevalences, group centers, singer variability, vocalization noise

z contains latent singer signatures

D_{train} contains observed vocalizations and training labels

What we obtain

$$\Theta^{(1)}, \Theta^{(2)}, \ldots, \Theta^{(M)}$$

Predictions are then computed by averaging class scores over these posterior draws:

$$S_c \approx \frac{1}{M} \sum_{m=1}^{M} p(c \mid \pi_{v_*}^{(m)}) p(\mathbf{X}_* \mid c, v_*, \Theta^{(m)})$$

Inference and Diagnostics

Implementation

We used PyMC for inference and ArviZ for model diagnostics

Checks

- Prior predictive sampling: checks whether the prior generates plausible standardized observations;
- diagnostics: MCMC diagnostics check whether independent chains provide a stable approximation of the posterior. Here the table shows the results obtained with these parameters :
    - seed = 2026
    - tune = 1000
    - target accept = 0.95
    - prior draws = 500

Diagnostics

Check	Value
Sampler	NUTS
Chains	4
Tune steps	1000
Posterior draws	1000 per chain
Divergences	0
Max R-hat	1.00
Min bulk ESS	2467
Min tail ESS	1476
Min BFMI	0.893

Evaluation and baselines

Singer-Disjoint Split

All vocalizations of a singer stay in the same split. Bootstrap resampling is done by singer.

Baselines

1

Prevalence

predict dramatic-class prevalence observed in the training set

2

Logistic Regression

aggregated features per singer

3

Random Forest

aggregated features per singer

Comparison

model	log loss	brier score	balanced accuracy	n singers
prevalence	0.688	0.247	0.5	20
logistic	0.579	0.198	0.687	20
random_forest	0.563	0.194	0.697	20
bayesian hierarchical	0.524	0.175	0.717	20

Bootstrap intervals are broad, which is consistent with the small number of held out singers

Results, limitations and takeaway

Uncertainty vs number of vocalizations

m vocalizations	Test singers	Mean P(true class)	Mean entropy
1	20	0.636	0.572
2	20	0.634	0.567
4	20	0.633	0.548
8	20	0.638	0.539
all	20	0.652	0.525

note: entropy is expressed in nats

Increasing the number of vocalizations gives only a weak reduction in uncertainty. The model remains cautious, which is consistent with a small held-out set and a reduced feature set

Honest limitations

- 20 held-out test singers
- Reduced public feature set: PHE + SC only
- Diagonal Gaussian; no full covariance
- Not a replication of the original paper's full classifier

Goal reached

Interpretable Bayesian singer-level model that separates singer variability from vocalization noise and returns posterior class probabilities for unseen singers.

Thank you for your attention!

you can find the code here: **https://github.com/tibebit/pml-project**